

TrishulHub AI Agent Transformation

Complete Master Plan

1. Executive Summary	4
2. AI Agents Your Company Needs	4
2.1 Dev Agent - The Code Builder	4
2.2 Client Hunter Agent - The Sales Machine	5
2.3 Finance Agent - The Money Manager	6
2.4 Project Manager Agent - The Task Organizer	6
2.5 HR Agent - The Team Coordinator	7
2.6 Content Agent - The Marketing Writer	7
2.7 Support Agent - The Client Helper	8
3. How Agents Work Together	8
4. Dashboard Design - Your Command Center	10
4.1 Login and User Roles	10
4.2 Main Dashboard Home Screen	10
4.3 Agent Control Panel	11
4.4 API Key Management (Critical Feature)	11
4.5 CRM (Client Relationship Manager)	12
4.6 Project Management	13
4.7 Finance Dashboard	13
4.8 Client Portal	13
4.9 Data Migration from Current App	14
5. Technology Stack and Cost Breakdown	14
5.1 Technology Choices	14

5.2 Monthly Cost Breakdown	15
6. AI Model Strategy - Getting Maximum Value	15
6.1 Tiered Model Routing	15
6.2 Cost-Saving Techniques	16
6.3 Estimated Monthly API Usage	16
7. Hostinger Setup Guide (Step-by-Step)	16
7.1 Step 1: Upgrade Your Hosting Plan	16
7.2 Step 2: Set Up Node.js Environment	17
7.3 Step 3: Set Up MySQL Database	17
7.4 Step 4: Connect Your GitHub Repository	17
7.5 Step 5: Point Your Domain	18
8. Step-by-Step Execution Plan	18
8.1 Phase 1: Foundation (Week 1-2)	18
8.2 Phase 2: Core Agent System (Week 3-4)	18
8.3 Phase 3: CRM and Client Acquisition (Week 5-6)	19
8.4 Phase 4: Finance and Project Management (Week 7-8)	19
8.5 Phase 5: Client Portal and Remaining Agents (Week 9-10)	20
8.6 Phase 6: Launch and Optimize (Week 11-12)	20
9. Cline Prompts - Building the System	21
9.1 Prompt for Phase 1 - Foundation	21
9.2 Prompt for Phase 2 - Core Agent System	22
9.3 Prompt for Phase 3 - CRM and Client Acquisition	22
9.4 Prompt for Phase 4 - Finance and Project Management	23
9.5 Prompt for Phase 5 - Client Portal and Remaining Agents	23
10. Important Notes and Tips	24
10.1 How to Use Cline Effectively	24

10.2 Security Considerations	25
10.3 Scaling the System	25
10.4 What If Something Goes Wrong?	25

1. Executive Summary

This master plan outlines a complete transformation of TrishulHub from a traditional web development SaaS company into an AI-agent-powered organization. The goal is simple but ambitious: every task in the company will be handled or assisted by AI agents, while your team members monitor, review, and approve the work. This means your company can serve more clients, deliver faster, and grow without needing to hire more people.

Currently, TrishulHub has 4 team members: Taroon (Founder and CEO), Pruthvi (Co-founder and BOD, focused on business development), Akshat (developer with basic technical knowledge using WordPress and PHP/HTML), and Kiran (developer with strong technical skills who built the current app). You serve 6 active clients and aim to acquire 4 new clients every month. Your current tools are a basic PHP/HTML management app at app.trishulhub.in and a Gmail account.

The new system will replace your current app with a powerful AI Agent Dashboard built in Next.js. This dashboard will be the central hub where your team gives tasks to AI agents, monitors their work, approves or rejects outputs, and manages clients. Clients will also get their own portal to view project progress, download invoices, and raise support tickets. The system is designed to work within your budget of approximately \$18 per month for API costs, using a smart mix of paid and free AI models through OpenRouter.

Key Outcome: TrishulHub will operate like a 20-person company with just 4 people, because AI agents will handle the heavy lifting in development, client acquisition, finance, HR, and project management.

2. AI Agents Your Company Needs

Based on deep research into your business model, team structure, and growth goals, I have designed 7 specialized AI agents. Each agent has a specific role, specific tasks it handles, and specific rules for when it needs human approval. Here is the complete breakdown:

2.1 Dev Agent - The Code Builder

This is the most important agent for your company. The Dev Agent writes code, builds websites and web apps, fixes bugs, creates templates, and deploys projects to Hostinger. It can understand project requirements in plain English and generate complete, working code. When Kiran or Akshat give it a task like "Build a 5-page business website for a restaurant client with a menu page, gallery, and contact form," it will generate all the HTML, CSS, JavaScript, and PHP files needed.

What Dev Agent Does Automatically:

- Writes complete website code in HTML, CSS, JavaScript, and PHP based on project requirements
- Creates responsive designs that work on mobile, tablet, and desktop
- Generates website templates for common business types (restaurants, portfolios, e-commerce, etc.)

- Fixes bugs when you describe the problem or upload a screenshot
- Writes deployment scripts for Hostinger
- Generates API documentation and code comments
- Creates database schemas and writes SQL queries for MySQL

What Needs Human Review (Kiran/Akshat):

- Final code review before deploying to production
- Architecture decisions for complex applications
- Security-sensitive code (payment integration, user authentication)
- Client-specific customizations that require creative judgment

AI Model Recommendation:

Use GPT-4o-mini for routine code generation (cheap at approximately \$0.15 per million input tokens). For complex applications or debugging, use DeepSeek R1 (free on OpenRouter) which is excellent at reasoning through code problems. For client-facing code that must be perfect, use GPT-4o (approximately \$2.50 per million tokens) but only when needed.

2.2 Client Hunter Agent - The Sales Machine

This is the agent that solves your biggest problem: finding new clients. Currently, you rely on friends and word of mouth, which is unpredictable. The Client Hunter Agent will actively search for businesses that need websites, reach out to them with personalized messages, and bring qualified leads to Pruthvi for closing. This agent works 24/7 and never gets tired of prospecting.

What Client Hunter Agent Does Automatically:

- Searches Google Maps, business directories, and social media for businesses with poor or no websites
- Scores leads based on how likely they are to need your services (1-100 score)
- Writes personalized cold emails referencing specific details about each prospect
- Sends follow-up emails on a schedule (day 3, day 7, day 14) if no response
- Posts content on Instagram and LinkedIn to attract inbound leads
- Tracks all leads in the CRM with status updates (New, Contacted, Interested, Negotiating, Won, Lost)
- Generates weekly reports on lead generation performance and conversion rates

What Needs Human Review (Pruthvi/Taroon):

- Every cold email must be approved before sending (to maintain quality and brand voice)
- Meeting scheduling and pricing negotiation
- Final proposal creation and sending

- Deciding which leads to pursue aggressively vs. which to deprioritize

Lead Generation Strategy:

The agent will use Apollo.io free tier (100 email credits per month) to find business leads. It will analyze each prospect's current website to identify weaknesses and opportunities. Personalized outreach emails will be sent via Resend.com (free: 100 emails per day). Social media content will be generated and scheduled automatically. The estimated cost for this entire pipeline is \$0 per month using free tiers, with the AI model costs covered by your existing budget.

2.3 Finance Agent - The Money Manager

The Finance Agent handles all money-related tasks so Taroon and Pruthvi can focus on growing the business instead of chasing invoices and tracking expenses. It generates invoices, sends payment reminders, tracks expenses, creates financial reports, and monitors cash flow. This agent ensures you always know exactly how much money is coming in and going out.

What Finance Agent Does Automatically:

- Generates professional invoices when a project milestone is completed
- Sends automatic payment reminders (3 days before due, on due date, 3 days after due)
- Tracks all income and expenses in a categorized ledger
- Creates monthly financial summary reports (revenue, expenses, profit, client-wise breakdown)
- Calculates project profitability (actual vs. estimated hours and costs)
- Tracks recurring costs (Hostinger, domain renewals, API costs)
- Generates GST/tax-ready reports for your accountant

What Needs Human Review (Taroon):

- Sending any invoice above a threshold amount (you decide the limit)
- Making payments or approving expenses above a set amount
- Final review of financial reports before sharing externally
- Any decision involving pricing changes or discounts

2.4 Project Manager Agent - The Task Organizer

The Project Manager Agent is like having a dedicated project manager who never forgets a deadline, never misses a follow-up, and always knows the status of every project. It creates tasks from client requests, assigns them to the right team member or AI agent, tracks progress, sends deadline reminders, and generates status reports. It ensures nothing falls through the cracks.

What Project Manager Agent Does Automatically:

- Creates projects and task breakdowns when a new client is onboarded
- Assigns tasks to Dev Agent for code work and to team members for review

- Sends deadline reminders (1 week, 3 days, 1 day before deadline)
- Tracks time spent on each task and project
- Generates weekly project status reports for all active projects
- Escalates blocked or delayed tasks to Taroon or Pruthvi
- Creates client-facing progress updates
- Converts meeting notes into action items with assigned owners and deadlines

What Needs Human Review (Any Team Member):

- Task priority decisions when multiple projects compete for resources
- Client communication about delays or scope changes
- Resource allocation decisions

2.5 HR Agent - The Team Coordinator

Even with a small team, HR tasks take time. The HR Agent handles attendance tracking, leave management, onboarding documents, team communication, and workload monitoring. It watches for signs that the team is overloaded and suggests when it might be time to consider additional help. As your company grows beyond 4 people, this agent becomes even more valuable.

What HR Agent Does Automatically:

- Tracks daily check-ins and working hours
- Manages leave requests and approvals (forwarded to Taroon for final approval)
- Creates onboarding documents for new team members
- Monitors workload per team member and alerts if someone is overloaded
- Sends daily stand-up summaries to the team
- Maintains a company knowledge base (processes, standards, templates)
- Generates monthly team productivity reports

What Needs Human Review (Taroon):

- Leave approvals
- Hiring decisions (the agent can suggest but not decide)
- Performance reviews and feedback

2.6 Content Agent - The Marketing Writer

The Content Agent creates all written content for your company: website copy for client projects, blog posts for SEO, social media posts for Instagram and LinkedIn, email templates, case studies, and portfolio descriptions. It works closely with the Client Hunter Agent to create outreach materials and with the Dev Agent to write website copy for client projects.

What Content Agent Does Automatically:

- Writes website copy for client projects based on their business type and requirements
- Creates social media posts for Instagram (images via AI image generation + captions)
- Writes blog posts about web development topics for SEO
- Generates case studies from completed projects
- Creates email templates for different scenarios (welcome, follow-up, thank you, etc.)
- Writes project proposals and quotations
- Creates portfolio descriptions for your website

What Needs Human Review (Pruthvi/Akshat):

- All client-facing content before publishing
- Brand voice consistency
- Technical accuracy of blog posts

2.7 Support Agent - The Client Helper

The Support Agent handles client support tickets, answers common questions, and provides first-level technical support. It can resolve simple issues automatically (like password resets, common how-to questions) and escalates complex issues to the right team member. Clients interact with this agent through the client portal.

What Support Agent Does Automatically:

- Answers common client questions (hosting details, domain info, how to access their site)
- Creates support tickets from client emails and portal submissions
- Categorizes and prioritizes tickets (urgent, high, medium, low)
- Resolves simple issues automatically (FAQ responses, basic troubleshooting)
- Routes complex issues to the right team member
- Sends client satisfaction surveys after ticket resolution
- Generates weekly support summary reports

What Needs Human Review (Kiran/Akshat):

- Any technical issue the agent cannot resolve
- Client complaints or dissatisfaction
- Requests for changes to live websites

3. How Agents Work Together

The agents do not work in isolation. They communicate with each other through a shared database and event system. Here is how a typical client journey flows through the system, from first contact to final delivery:

Step-by-Step Client Journey:

Step 1 - Lead Discovery: Client Hunter Agent finds a business with an outdated website and creates a lead in the CRM.

Step 2 - Outreach: Client Hunter Agent drafts a personalized email. Pruthvi reviews and approves it. The email is sent.

Step 3 - Qualification: If the prospect responds positively, Client Hunter Agent creates a qualified lead and notifies Pruthvi.

Step 4 - Proposal: Content Agent generates a project proposal. Pruthvi customizes and sends it.

Step 5 - Project Creation: When the client says yes, Project Manager Agent creates the project with tasks and deadlines.

Step 6 - Development: Dev Agent writes the code. Kiran reviews and provides feedback via screenshots. Dev Agent revises.

Step 7 - Content: Content Agent writes website copy. Akshat reviews and approves.

Step 8 - Deployment: Dev Agent deploys to Hostinger. Kiran does final review and approves.

Step 9 - Invoicing: Finance Agent generates the invoice. Taroon reviews and approves. Invoice sent to client.

Step 10 - Support: Support Agent handles ongoing client questions. Complex issues go to Kiran or Akshat.

Agent-to-Team-Member Responsibility Map:

Agent	Primary Monitor	Interaction Mode	Approval Required
Dev Agent	Kiran	AI writes code, Kiran reviews and approves/rejects	Yes - all outputs
Client Hunter	Pruthvi	AI finds leads and drafts emails, Pruthvi approves	Yes - all outreach
Finance Agent	Taroon	AI prepares invoices and reports, Taroon approves	Yes - all payments
Project Manager	All Team	AI organizes and tracks, team reviews status	For priority changes
HR Agent	Taroon	AI tracks and suggests, Taroon decides	For leave/hiring
Content Agent	Pruthvi/Akshat	AI writes, team reviews and edits	Yes - client content

Support Agent	Kiran/Akshat	AI handles simple issues, team handles complex	For escalations
---------------	--------------	--	-----------------

4. Dashboard Design - Your Command Center

The AI Agent Dashboard will be a web application accessible at app.trishulhub.in (replacing your current app). It will be built using Next.js and hosted on Hostinger. Every team member and client will have their own login. Here is the complete breakdown of every screen and feature the dashboard must have:

4.1 Login and User Roles

The dashboard will have 4 user roles, each with different permissions. When someone logs in, they see only the features relevant to their role. This keeps the interface clean and secure.

Role	Who	Access Level	Can Do
Super Admin	Taroon	Full Access	Everything + settings, billing, team management, API keys
Admin	Pruthvi	Most Access	CRM, projects, finance view, client communication, approve outreach
Developer	Kiran, Akshat	Project Access	Dev tasks, code review, deploy, support tickets, give feedback to AI
Client	Clients	Limited Access	View project progress, download invoices, raise tickets, see who worked on project

4.2 Main Dashboard Home Screen

When a team member logs in, they see the main dashboard. This is the nerve center of the entire system. It shows a snapshot of everything happening in the company at a glance, with quick actions for the most common tasks.

Widgets on the Home Screen:

Agent Status Panel: Shows all 7 agents with their current status (Running, Idle, Waiting for Approval, Error). Each agent shows a colored dot: green = running, gray = idle, yellow = waiting for approval, red = error.

Pending Approvals Queue: A list of tasks waiting for YOUR approval. Each item shows the agent name, task description, and Approve/Reject/Edit buttons. This is where you review AI outputs before they are finalized.

Active Projects Summary: Cards showing each active project with progress percentage, next deadline, and assigned team members.

Revenue This Month: Total income, pending invoices, and overdue payments at a glance.

New Leads Today: Number of new leads found by Client Hunter Agent today, with a quick view button.

API Usage Tracker: Shows how much of your API budget has been used this month, with a progress bar and remaining dollar amount. Broken down by agent.

Quick Action Buttons: "Create New Project", "Give Task to Agent", "Send Invoice", "View Reports".

4.3 Agent Control Panel

This is where you interact with each AI agent. You can give tasks, review outputs, provide feedback, and manage settings. Each agent has its own dedicated page within this panel.

Features on Each Agent Page:

Task Input Box: A text area where you type what you want the agent to do in plain English. For the Dev Agent, you might type "Build a landing page for a dental clinic with appointment booking form." For the Content Agent, you might type "Write 3 Instagram captions about our new website design service."

File/Screenshot Upload: A drag-and-drop area to upload files (images, documents, code files) or screenshots. This is essential for giving feedback to agents. For example, if the Dev Agent built a page but the layout is wrong, Akshat can take a screenshot, circle the problem area, upload it, and type "Fix the header alignment as shown in this screenshot."

Output Viewer: Shows the agent's latest output. For code, it shows a code preview. For emails, it shows the email text. For invoices, it shows the invoice preview. For websites, it can show a live preview link.

Approve/Reject/Edit Buttons: After reviewing the output, click Approve to finalize, Reject to discard, or Edit to modify before approving.

Conversation History: A chat-like view showing all interactions with this agent, including your instructions, agent responses, feedback, and revisions.

Agent Settings: Configure which AI model this agent uses, set approval requirements, and adjust behavior preferences.

4.4 API Key Management (Critical Feature)

This is one of the most important features you requested. Since you have a limited budget for AI APIs, you need to be able to manage multiple API keys, see how much limit is left on each, and switch keys when one runs out. The API Key Management page gives you complete control.

How It Works:

You can add multiple API keys from different providers. Each key is assigned to one or more agents. When an agent needs to call an AI model, the system checks which key to use based on the agent assignment and current usage. If a key runs out of credits or hits its limit, the system automatically switches to the next

available key. If no keys are available, it alerts you and pauses the agent until you add a new key.

API Key Dashboard Shows:

Key List: All added API keys with provider name (OpenRouter, z.ai, etc.), key nickname, status (Active/Exhausted/Error), and assigned agents.

Usage Per Key: How many tokens used this month, estimated cost, and remaining balance. A progress bar shows percentage of budget used.

Usage Per Agent: How much each agent is consuming. This helps you identify if one agent is burning through your budget too fast.

Auto-Failover: If Key A runs out, automatically switch to Key B. You set the priority order.

Budget Alerts: Yellow warning at 50% of monthly budget, orange at 75%, red at 90%. Email notification at each threshold.

Model Downgrade: When budget hits 80%, automatically switch agents to cheaper or free models (like from GPT-4o to GPT-4o-mini to free Llama 3.3 70B on OpenRouter).

Provider	What You Get	Cost	Best For
z.ai API	\$18/month plan - multiple model access	\$18/month	Primary - all agents
OpenRouter	Access to 300+ models including free ones	Free tier available	Backup + free models
OpenRouter Free Models	DeepSeek R1, Llama 3.3 70B, GLM, Qwen 2.5	\$0	Routine tasks, overflow

4.5 CRM (Client Relationship Manager)

The CRM replaces the basic client management in your current app. It is powered by the Client Hunter Agent for lead generation and gives Pruthvi and Taroon full visibility into the sales pipeline.

CRM Features:

Lead Pipeline: Kanban board view with columns: New Lead, Contacted, Interested, Proposal Sent, Negotiating, Won, Lost. Drag leads between columns.

Lead Details: Click any lead to see company name, contact person, email, phone, website URL, current website quality score, AI-generated outreach suggestions, and full communication history.

Auto-Lead Import: Client Hunter Agent automatically adds new leads. You can also manually add leads.

Email Integration: Send emails directly from the CRM. All replies are tracked and logged automatically.

Follow-up Reminders: The system reminds you when a follow-up is due. AI suggests what to say based on previous interactions.

Conversion Analytics: See how many leads convert at each stage, average time to close, and which lead sources perform best.

4.6 Project Management

This module manages all client projects from creation to delivery. It replaces the basic project tracking in your current app with a much more powerful system powered by the Project Manager Agent.

Project Management Features:

Project Board: Kanban view with columns: Planning, In Progress, Review, Client Approval, Deployed, Completed.

Task Management: Each project has tasks with assignee (team member or AI agent), deadline, priority, status, and file attachments.

AI Task Assignment: When you create a project, the Project Manager Agent suggests a task breakdown with estimated timelines.

Time Tracking: Automatic time logging when work is done. AI agents log their processing time. Team members log their review time.

File Management: Upload project files, code, designs, and documents. Everything is organized by project.

Client Updates: Automatically send progress updates to clients at configurable intervals (daily, weekly, or on milestone completion).

4.7 Finance Dashboard

The Finance Dashboard gives Taroon complete visibility into the company's financial health. The Finance Agent handles the number-crunching, and Taroon just reviews and approves.

Finance Dashboard Features:

Revenue Overview: This month's revenue, pending payments, overdue invoices, and revenue trend chart (last 6 months).

Invoice Manager: Create, send, and track invoices. AI generates invoice from project data. One-click send to client.

Expense Tracker: Log expenses with categories (Hosting, Domains, API Costs, Tools, Other). Receipt upload supported.

Profit and Loss Report: Auto-generated monthly P&L; statement.

Client Payment History: See all payments from each client with dates, amounts, and invoice numbers.

Budget Tracker: Compare actual spending vs. budget. Alerts when overspending.

4.8 Client Portal

Clients get their own login to the system. This is a simplified view that shows only information relevant to them. It replaces and improves upon the client features in your current app.

Client Portal Features:

Project Dashboard: See all their projects with current status, progress percentage, and next milestone.

Team Activity: See who worked on their project last (like your current app), with timestamps and task descriptions.

Invoice Center: View all invoices, download as PDF, see payment status (Paid, Pending, Overdue).

Support Tickets: Raise new tickets, view existing tickets and their status, communicate with support team. This is NEW compared to your current app.

File Downloads: Access project deliverables, source files, and documents.

Feedback: Give feedback on completed work with text and screenshot attachments.

4.9 Data Migration from Current App

Your current app.trishulhub.in has existing data that needs to be moved to the new system. The migration plan is as follows:

Client Data: All client names, emails, phone numbers, and project details will be imported into the new CRM.

Project Data: All active and completed projects will be imported with their status, deadlines, and assigned team members.

Invoice Data: All past invoices will be imported into the Finance Dashboard for historical records.

Login Credentials: All existing client login credentials will be migrated so clients do not need to re-register.

You will need to export your current data from the old app (as CSV or SQL dump). During the Cline setup process, Kiran can help with this. The new system will have an import tool that accepts CSV files for easy data loading.

5. Technology Stack and Cost Breakdown

5.1 Technology Choices

Component	Technology	Why
Frontend + Backend	Next.js 16	Modern framework, handles both UI and API, best for AI integration
Database	MySQL	Free on Hostinger, well-supported, reliable for business data

ORM	Prisma	Makes database operations easy and safe, auto-generates queries
UI Components	shadcn/ui + Tailwind CSS	Professional-looking interface, fast to build, fully customizable
AI Gateway	OpenRouter	One API for 300+ models, includes free models, easy key management
Agent Framework	Custom (Next.js API)	Built directly into your app, no external framework needed, full control
Authentication	NextAuth.js	Secure login for team and clients, supports role-based access
Email Service	Resend (free tier)	100 emails per day free, perfect for outreach and notifications
File Storage	Hostinger + Local	Store uploads on your server, no extra cost
Hosting	Hostinger Cloud	Supports Node.js, MySQL included, affordable upgrade

5.2 Monthly Cost Breakdown

Item	Cost	Notes
Hostinger Cloud Hosting	\$7.99 - \$12.99/month	Upgrade from current plan. Includes MySQL, Node.js support, SSL
z.ai API Plan	\$18/month	Your current plan - primary AI model access
OpenRouter Free Models	\$0	DeepSeek R1, Llama 3.3 70B, GLM - backup models
Domain (trishulhub.in)	Already paid	No additional cost
Email (Resend)	\$0 (free tier)	100 emails/day - enough for current volume
Lead Finder (Apollo.io)	\$0 (free tier)	100 email credits/month - good starting point
Total Monthly Cost	\$26 - \$31/month	Well within budget when API costs are optimized

6. AI Model Strategy - Getting Maximum Value

Your biggest concern is the \$18/month API budget. The key to making this work is using the right AI model for each task. Not every task needs the most expensive model. Here is the strategy that will keep your costs low while maintaining high quality:

6.1 Tiered Model Routing

Task Type	Model	Cost	Used By
-----------	-------	------	---------

Simple tasks (formatting, categorization, FAQ answers)	Llama 3.3 70B (free on OpenRouter)	\$0	Support, HR
Medium tasks (email drafts, content writing, code generation)	GPT-4o-mini via OpenRouter	~\$0.15/M tokens	Content, Dev, Sales
Complex tasks (reasoning, debugging, analysis)	DeepSeek R1 (free on OpenRouter)	\$0	Dev, Finance
Critical tasks (client-facing code, important emails)	GPT-4o or Claude Sonnet	~\$2.50/M tokens	Dev (rare)

6.2 Cost-Saving Techniques

- Smart Caching:** The system remembers previous AI responses. If someone asks a similar question again, it uses the cached answer instead of calling the API again. This can save 30-80% on repeated queries.
- Response Length Limits:** Set maximum response lengths for each task type. A 2-sentence email draft does not need a 500-word response. This saves tokens on every call.
- Auto-Downgrade:** When your monthly budget hits 80%, the system automatically switches all agents to free models (Llama, DeepSeek R1). Quality drops slightly but work continues without interruption.
- Batch Processing:** Group similar tasks together and process them in one API call instead of multiple calls. For example, generate 5 social media posts in one call instead of 5 separate calls.
- Context Summarization:** Instead of sending full conversation history every time, summarize previous context. A 1000-token summary replaces a 5000-token history, saving 80% on input tokens.

6.3 Estimated Monthly API Usage

Usage Level	Agent Runs/Day	Model Mix	Est. Monthly Cost
Light (starting phase)	~50	80% free + 20% paid	\$3 - \$8
Medium (growth phase)	~200	60% free + 40% paid	\$8 - \$18
Heavy (scaling phase)	~500+	40% free + 60% paid	\$18 - \$40

7. Hostinger Setup Guide (Step-by-Step)

You mentioned you are not from a tech background and currently have a Premium Web Hosting plan on Hostinger that only supports PHP/HTML. You need to upgrade to a plan that supports Node.js. Here is exactly what to do, explained in simple terms:

7.1 Step 1: Upgrade Your Hosting Plan

Log in to Hostinger: Go to hostinger.com and log in to your account.

Go to Hosting Plans: Click on "Hosting" in the top menu, then click "Upgrade" on your current plan.

Choose Cloud Startup Plan: This costs approximately \$7.99/month (promotional price, renews at \$25.99/month). It supports Node.js applications, includes unlimited MySQL databases, 3GB RAM, and can handle up to 10 Node.js apps. This is the minimum plan you need.

Complete the Purchase: Pay for the upgrade. Your existing website and data will NOT be affected during the upgrade.

Wait for Activation: The upgrade usually takes 15-30 minutes. You will get an email confirmation.

7.2 Step 2: Set Up Node.js Environment

Open hPanel: Go to your Hostinger dashboard (hPanel).

Find Node.js Section: Look for "Node.js" or "Advanced" in the left sidebar menu. Click on it.

Create Node.js App: Click "Create Application". Set the App Name to "trishulhub-dashboard". Select Node.js version 20.x (the latest stable version). Set your domain (app.trishulhub.in) as the app URL.

Set Environment Variables: In the app settings, add the following environment variables (these will be provided by Kiran or generated during Cline setup): DATABASE_URL, OPENROUTER_API_KEY, ZAI_API_KEY, NEXTAUTH_SECRET, and NEXTAUTH_URL.

7.3 Step 3: Set Up MySQL Database

Go to Databases: In hPanel, click "MySQL Databases" in the left sidebar.

Create Database: Click "Create New Database". Name it something like "trishulhub_main". Create a database user with a strong password. Note down the database name, username, and password - you will need these.

Access phpMyAdmin: You can use phpMyAdmin (available in hPanel) to view and manage your database data. This is useful for checking if the migration worked correctly.

7.4 Step 4: Connect Your GitHub Repository

Create GitHub Account: If you do not have one, create a free account at github.com.

Create Repository: Create a new repository called "trishulhub-dashboard". Make it private so your code is secure.

Connect to Hostinger: In hPanel, under your Node.js app settings, there is a "Deployment" section. Connect your GitHub repository. Set the branch to "main", build command to "npm run build", and start command to "npm run start".

Auto-Deploy: Once connected, every time you push code to GitHub (which Cline will do automatically), Hostinger will rebuild and redeploy your app. This usually takes 2-5 minutes.

7.5 Step 5: Point Your Domain

If app.trishulhub.in was already pointing to your old app, you need to update the DNS or the Node.js app settings to serve the new dashboard. In most cases, Hostinger handles this automatically when you set the app URL during Node.js app creation. If the old app was in a subdirectory, Kiran can help redirect the subdomain to the new Node.js app.

8. Step-by-Step Execution Plan

This is the complete execution plan. Each phase has clear steps, who is responsible, and what the expected output is. Follow this plan sequentially - do not skip steps. The entire build can be done using Cline in VS Code, with Kiran monitoring and reviewing the generated code.

8.1 Phase 1: Foundation (Week 1-2)

Goal: Set up the hosting environment, create the project structure, and build the basic dashboard with login and user roles.

Step	Task	Who	Output
1.1	Upgrade Hostinger plan to Cloud Startup	Taroon	Node.js hosting ready
1.2	Create MySQL database on Hostinger	Taroon	Database credentials saved
1.3	Create GitHub repository "trishulhub-dashboard"	Kiran	Private repo created
1.4	Initialize Next.js project with TypeScript, Tailwind, Prisma	Cline + Kiran	Project skeleton ready
1.5	Set up database schema (users, projects, clients, invoices, etc.)	Cline + Kiran	All tables created
1.6	Build authentication (login, register, password reset) with NextAuth.js	Cline + Kiran	Working login system
1.7	Create user role system (Super Admin, Admin, Developer, Client)	Cline + Kiran	Role-based access working
1.8	Build basic dashboard layout with sidebar navigation	Cline + Kiran	Dashboard shell ready
1.9	Deploy to Hostinger and verify it works	Kiran	Live app at app.trishulhub.in

8.2 Phase 2: Core Agent System (Week 3-4)

Goal: Build the AI agent infrastructure, API key management, and the Dev Agent as the first working agent.

Step	Task	Who	Output
2.1	Build OpenRouter API integration module	Cline + Kiran	API calls working
2.2	Build API Key Management page (add/edit/delete keys, assign to agents)	Cline + Kiran	API key dashboard working
2.3	Build API Usage Tracker (per-key and per-agent usage, budget alerts)	Cline + Kiran	Usage tracking visible
2.4	Build Agent Control Panel (task input, output viewer, approve/reject)	Cline + Kiran	Agent UI ready
2.5	Build file/screenshot upload feature for agent feedback	Cline + Kiran	File upload working
2.6	Build Dev Agent with code generation capability	Cline + Kiran	Dev Agent generates code
2.7	Test Dev Agent: give it a sample project and review output	Kiran + Akshat	Dev Agent tested and working
2.8	Build auto-failover system for API keys	Cline + Kiran	Key switching works

8.3 Phase 3: CRM and Client Acquisition (Week 5-6)

Goal: Build the CRM module and Client Hunter Agent so you can start finding new clients automatically.

Step	Task	Who	Output
3.1	Build CRM with lead pipeline (Kanban board), lead details, and status tracking	Cline + Kiran	CRM working
3.2	Build Client Hunter Agent with lead finding and email drafting	Cline + Kiran	Agent finds leads
3.3	Integrate Resend.com for sending emails from the CRM	Cline + Kiran	Email sending works
3.4	Build email approval queue for Pruthvi	Cline + Kiran	Approval workflow ready
3.5	Build follow-up automation system	Cline + Kiran	Auto follow-ups work
3.6	Test Client Hunter: find 10 sample leads and draft outreach emails	Pruthvi	Agent tested and working

8.4 Phase 4: Finance and Project Management (Week 7-8)

Goal: Build the Finance Dashboard, Project Management module, and their associated agents.

Step	Task	Who	Output
------	------	-----	--------

4.1	Build invoice generation and management system	Cline + Kiran	Invoices working
4.2	Build Finance Agent with invoice auto-generation and payment tracking	Cline + Kiran	Finance Agent working
4.3	Build expense tracker and financial reports	Cline + Kiran	Reports generating
4.4	Build Project Management module with Kanban board and task tracking	Cline + Kiran	PM module working
4.5	Build Project Manager Agent with task breakdown and deadline tracking	Cline + Kiran	PM Agent working
4.6	Build invoice approval workflow for Taroon	Cline + Kiran	Approval working
4.7	Migrate existing data from old app (clients, projects, invoices)	Kiran + Cline	Data migrated successfully

8.5 Phase 5: Client Portal and Remaining Agents (Week 9-10)

Goal: Build the client portal, Support Agent, Content Agent, and HR Agent.

Step	Task	Who	Output
5.1	Build Client Portal (project view, invoice download, support tickets)	Cline + Kiran	Client portal working
5.2	Build Support Agent with ticket handling and auto-responses	Cline + Kiran	Support Agent working
5.3	Build Content Agent with content generation for social media and websites	Cline + Kiran	Content Agent working
5.4	Build HR Agent with attendance, leave, and workload tracking	Cline + Kiran	HR Agent working
5.5	Migrate client login credentials to new system	Kiran	Clients can log in
5.6	Full system testing with all agents	All Team	Everything tested

8.6 Phase 6: Launch and Optimize (Week 11-12)

Goal: Go live with the new system, train the team, and start using AI agents for real work.

Step	Task	Who	Output
6.1	Switch app.trishulhub.in DNS to new dashboard	Kiran	New app is live
6.2	Team training session: how to use the dashboard and give tasks to agents	Kiran + Taroon	Team knows the system

6.3	Start using Dev Agent for a real client project	Kiran + Akshat	First AI-assisted project
6.4	Activate Client Hunter Agent to start finding leads	Pruthvi	Lead generation active
6.5	Monitor API usage for first 2 weeks and optimize model routing	Taroon + Kiran	Costs optimized
6.6	Gather team feedback and prioritize improvements	All Team	Improvement list ready

9. Cline Prompts - Building the System

These are the prompts you will paste into Cline in VS Code to build each phase. Cline will write the code, and Kiran will review and commit it to GitHub. Hostinger will automatically deploy. Open each prompt block, copy the entire text, and paste it into Cline. Wait for Cline to finish before moving to the next prompt.

Important: Before starting, make sure you have: (1) VS Code installed with the Cline extension, (2) Git installed and logged into your GitHub account, (3) Your Hostinger Cloud plan upgraded and MySQL database created, (4) Your OpenRouter API key ready (get one free at openrouter.ai).

9.1 Prompt for Phase 1 - Foundation

Prompt (copy everything below the line):

Build a Next.js 16 web application called "TrishulHub AI Dashboard". This is an AI agent management platform for a web development SaaS company. Here are the complete requirements: **TECH STACK:** Next.js 16 with App Router, TypeScript, Tailwind CSS 4, shadcn/ui components, Prisma ORM with MySQL, NextAuth.js for authentication. **DATABASE SCHEMA:** Create these tables - (1) User: id, name, email, password, role (SUPER_ADMIN/ADMIN/DEVELOPER/CLIENT), avatar, createdAt. (2) ApiKey: id, provider (OPENROUTER/ZAI/OTHER), keyName, keyValue (encrypted), monthlyBudget, currentSpend, status (ACTIVE/EXHAUSTED/ERROR), priority, assignedAgents (JSON array), createdAt. (3) Agent: id, name, type (DEV/CLIENT_HUNTER/FINANCE/PROJECT_MANAGER/HR/CONTENT/SUPPORT), description, model, systemPrompt, status, apiKeyId, createdAt. (4) Client: id, name, email, phone, company, website, status (ACTIVE/INACTIVE), userId, createdAt. (5) Project: id, name, clientId, status (PLANNING/IN_PROGRESS/REVIEW/APPROVAL/DEPLOYED/COMPLETED), progress, deadline, createdAt. (6) Task: id, title, description, projectId, assignedTo (userId or agentId), type (HUMAN/AI), status, priority, deadline, createdAt. (7) Invoice: id, invoiceNumber, clientId, projectId, amount, status (DRAFT/SENT/PAID/OVERDUE), dueDate, createdAt. (8) Lead: id, name, email, company, website, source, score, status (NEW/CONTACTED/INTERESTED/PROPOSAL/NEGOTIATING/WON/LOST), notes, createdAt. (9) AgentConversation: id, agentId, userId, messages (JSON), status, createdAt. (10) ApiUsageLog: id, apiKeyId, agentId, model, inputTokens, outputTokens, cost, createdAt. **AUTHENTICATION:** Use NextAuth.js with credentials provider. Create default users on seed: taroon@trishulhub.in (Super Admin), pruthvi@trishulhub.in (Admin), kiran@trishulhub.in (Developer), akshat@trishulhub.in (Developer). **LAYOUT:** Dashboard with left sidebar navigation. Sidebar items depend on role. Super Admin sees: Dashboard, Agents, CRM, Projects, Finance, Team, API Keys, Settings. Admin sees: Dashboard, CRM, Projects, Finance, Settings.

Developer sees: Dashboard, Projects (assigned), Agent Chat, Settings. Client sees: My Projects, Invoices, Support, Settings. DASHBOARD HOME: Show widgets - Agent Status Panel (all 7 agents with colored status dots), Pending Approvals Queue (list with approve/reject buttons), Active Projects Summary (cards with progress), Revenue This Month, New Leads Today, API Usage Tracker (progress bar with remaining budget), and Quick Action Buttons. Make the UI professional with shadcn/ui components. Use a clean, modern design with the TrishulHub branding. Configure for MySQL database with the DATABASE_URL environment variable. Set NEXTAUTH_URL and NEXTAUTH_SECRET. Create a .env.example file with all required variables. After creating the project, push it to GitHub repository "trishulhub-dashboard".

9.2 Prompt for Phase 2 - Core Agent System

Prompt (copy everything below the line):

Continue building the TrishulHub AI Dashboard. Now add the AI agent system with these features: OPENROUTER INTEGRATION: Create an API module at /lib/ai/openrouter.ts that handles all AI model calls through OpenRouter API. It should support: (1) Sending messages to any model available on OpenRouter, (2) Streaming responses, (3) Tracking token usage per call, (4) Automatic retry with a different model if the primary fails, (5) Support for vision/multimodal models (for screenshot analysis). Use the endpoint <https://openrouter.ai/api/v1/chat/completions>. API KEY MANAGEMENT PAGE: Build at /dashboard/api-keys. Features: (1) Add new API key form (provider, key name, key value, monthly budget, assigned agents), (2) Key list with status indicators (green=active, red=exhausted), (3) Usage per key (tokens used, cost, remaining budget with progress bar), (4) Usage per agent breakdown, (5) Auto-failover: when a key hits its budget limit, automatically switch to next priority key, (6) Budget alerts: yellow at 50%, orange at 75%, red at 90%, (7) Auto-downgrade: at 80% budget, switch all agents to free models (meta-llama/llama-3.3-70b-instruct:free on OpenRouter). AGENT CONTROL PANEL: Build at /dashboard/agents/[agentId]. Features: (1) Task input box - large text area where user types instructions in plain English, (2) File/screenshot upload - drag-and-drop area supporting images and files, uploaded files are stored locally and their URLs are passed to the AI model, (3) Output viewer - displays agent response (formatted text, code blocks with syntax highlighting, or preview), (4) Approve/Reject/Edit buttons below output, (5) Conversation history - chat-like view of all messages between user and agent, (6) Agent settings panel - configure model, system prompt, approval requirements. DEV AGENT: Create the first working agent. System prompt: "You are an expert web developer. You write clean, responsive HTML, CSS, JavaScript, and PHP code. When given a project requirement, you generate complete, working code. When given a screenshot with feedback, you analyze the image and fix the issues. Always include comments in your code." Default model: openai/gpt-4o-mini. When user uploads a screenshot, switch to a vision model (openai/gpt-4o or meta-llama/llama-3.2-11b-vision-instruct:free). Store all conversations in AgentConversation table. Log all API calls in ApiUsageLog table. API USAGE TRACKER WIDGET: Add to dashboard home. Shows: (1) Total budget, (2) Amount spent this month, (3) Breakdown by agent, (4) Breakdown by model, (5) Daily usage chart for last 30 days, (6) Projected end-of-month spend based on current trajectory.

9.3 Prompt for Phase 3 - CRM and Client Acquisition

Prompt (copy everything below the line):

Continue building the TrishulHub AI Dashboard. Now add the CRM module and Client Hunter Agent: CRM MODULE at /dashboard/crm: (1) Kanban board with columns: New Lead, Contacted, Interested, Proposal Sent, Negotiating, Won, Lost. Drag leads between columns. (2) Lead detail page showing: company name, contact

person, email, phone, website, current website quality score (1-10), outreach suggestions from AI, full communication history. (3) Add lead manually form. (4) Import leads from CSV. (5) Lead source tracking (Manual, AI Found, Referral, Social Media). (6) Conversion analytics page: conversion rate at each stage, average time to close, best performing lead sources, monthly trend chart. CLIENT HUNTER AGENT: System prompt: "You are an expert sales agent for a web development company called TrishulHub. Your job is to find businesses that need websites and reach out to them. When given a type of business or location, you suggest lead criteria. When given a lead, you write personalized cold emails that reference specific details about their business. Keep emails short, professional, and focused on the value TrishulHub provides." Features: (1) Lead search: user specifies business type and/or location, agent generates a list of search strategies and suggested lead criteria, (2) Email drafting: given a lead, agent drafts a personalized cold email, (3) Follow-up drafting: given a previous email and no response, draft a follow-up, (4) All emails go to approval queue before sending. EMAIL INTEGRATION: Integrate Resend.com API for sending emails. Create email templates. All emails are logged in the CRM under the lead. Track email opens and clicks via webhooks. APPROVAL QUEUE: Build a dedicated page at /dashboard/approvals. Shows all pending items that need human review. Each item shows: agent name, task type, preview of the output, and Approve/Reject/Edit buttons. Approved items are executed (emails sent, invoices delivered, code committed). Rejected items return to the agent with rejection reason. Edited items allow the user to modify before approving.

9.4 Prompt for Phase 4 - Finance and Project Management

Prompt (copy everything below the line):

Continue building the TrishulHub AI Dashboard. Now add Finance Dashboard and Project Management: FINANCE DASHBOARD at /dashboard/finance: (1) Revenue overview: this month revenue, pending payments, overdue invoices, revenue trend chart for last 6 months. (2) Invoice manager: create invoice form (select client, project, line items, tax, due date), preview invoice as PDF, send invoice to client email, mark as paid. (3) Expense tracker: log expense form (amount, category, date, receipt upload), expense categories: Hosting, Domains, API Costs, Tools, Marketing, Other. (4) Profit and Loss report: auto-generated monthly, shows total income, total expenses, net profit, with charts. (5) Client payment history: view all payments from each client with dates, amounts, invoice numbers. (6) Budget tracker: compare actual vs budget, alert when overspending. FINANCE AGENT: System prompt: "You are a financial assistant for TrishulHub. You generate professional invoices, track payments, create financial reports, and send payment reminders. Always calculate amounts accurately and format financial data clearly." Features: (1) Auto-generate invoice from project milestone data, (2) Payment reminder automation (3 days before due, on due date, 3 days after due), (3) Monthly financial summary report generation. All invoices above a configurable threshold require Taroan approval. PROJECT MANAGEMENT at /dashboard/projects: (1) Project list with status filters, (2) Project detail page with Kanban task board, (3) Task creation form with assignee (team member or AI agent), priority, deadline, (4) Time tracking per task, (5) File attachments per project, (6) Client update notifications. PROJECT MANAGER AGENT: System prompt: "You are a project manager for TrishulHub. You break down projects into tasks, set realistic deadlines, track progress, and ensure nothing is missed. When given a project description, you create a detailed task breakdown with estimated hours and dependencies." Features: (1) Task breakdown from project description, (2) Deadline reminders (1 week, 3 days, 1 day before), (3) Weekly status report generation, (4) Escalation alerts for blocked or delayed tasks. DATA MIGRATION: Create a CSV import tool at /dashboard/settings/import. Accept CSV files for clients, projects, and invoices. Map CSV columns to database fields. Validate data before import. Show import summary with count of records imported, skipped, and errors.

9.5 Prompt for Phase 5 - Client Portal and Remaining Agents

Prompt (copy everything below the line):

Continue building the TrishulHub AI Dashboard. Now add Client Portal and remaining agents: CLIENT PORTAL at /portal (separate layout from admin dashboard): (1) Client dashboard: show their projects with status, progress percentage, next milestone. (2) Project detail: see who worked last (team member name and timestamp), task list with statuses, project timeline. (3) Invoice center: list all invoices, download as PDF, see payment status (Paid/Pending/Overdue). (4) Support tickets: raise new ticket (text + file attachment), view existing tickets and status, reply to tickets. (5) Feedback: give feedback on completed work with text and screenshot attachments. (6) Client profile: update contact info and password. SUPPORT AGENT: System prompt: "You are a customer support agent for TrishulHub. You help clients with their questions about their websites, hosting, domains, and general issues. You are friendly, patient, and thorough. If you cannot resolve an issue, you escalate it to a human team member." Features: (1) Auto-respond to common FAQs (hosting details, domain info, how to access site), (2) Create tickets from client submissions, (3) Categorize and prioritize tickets (urgent/high/medium/low), (4) Auto-resolve simple issues, (5) Escalate complex issues to Kiran or Akshat, (6) Client satisfaction survey after ticket resolution. CONTENT AGENT: System prompt: "You are a content writer for TrishulHub. You write website copy, social media posts, blog articles, email templates, and case studies. Your writing is professional, engaging, and optimized for SEO. You adapt your tone based on the platform and audience." Features: (1) Website copy generation for client projects, (2) Social media post generation (Instagram captions, LinkedIn posts), (3) Blog post writing for SEO, (4) Case study creation from completed projects, (5) Project proposal and quotation drafting. All client-facing content requires approval. HR AGENT: System prompt: "You are an HR coordinator for TrishulHub. You track attendance, manage leave requests, monitor workload, and help with team coordination. You are organized and proactive about alerting management to potential issues." Features: (1) Daily check-in tracking, (2) Leave request management (submit to Taroon for approval), (3) Workload monitoring and overload alerts, (4) Daily stand-up summary, (5) Monthly productivity report generation. Also add: Notification system (in-app bell icon + email for important events), dark mode toggle, and mobile responsive design for all pages.

10. Important Notes and Tips

10.1 How to Use Cline Effectively

Cline is an AI coding assistant that runs inside VS Code. It reads your instructions and writes code automatically. Here are some tips for getting the best results:

One prompt at a time: Do not paste all 5 prompts at once. Complete one phase, test it, fix any issues, then move to the next prompt. This ensures each layer is solid before building on top of it.

Be specific about errors: If Cline makes a mistake or something does not work, tell it exactly what went wrong. For example, "The login page shows a blank screen" is better than "It does not work."

Let Kiran review: After each phase, have Kiran review the generated code. He can fix issues, improve structure, and ensure security. Cline is good but not perfect - human review is essential.

Commit frequently: After each successful phase, commit the code to GitHub. This creates a backup. If something goes wrong in the next phase, you can always go back to the last working version.

Use .env file: Never put API keys or passwords directly in the code. Always use environment variables. The .env.example file will show you what variables are needed.

10.2 Security Considerations

API Keys: Store all API keys encrypted in the database. Never expose them in the frontend code.

Passwords: Use bcrypt hashing for all user passwords. NextAuth.js handles this automatically.

Client Data: Clients can only see their own projects and invoices. Never expose other clients' data.

Agent Actions: All critical agent actions (sending emails, deploying code, sending invoices) require human approval. The system should never take irreversible action without consent.

HTTPS: Hostinger provides free SSL certificates. Make sure your app is always accessed over HTTPS.

10.3 Scaling the System

As TrishulHub grows from 6 clients to 20, 50, or 100 clients, the system will need to scale. Here is how to handle growth:

More API Budget: As client volume increases, your API costs will grow. When you consistently hit the \$18/month limit, consider upgrading to a \$50 or \$100/month API plan. The system's auto-downgrade feature will keep things running even on a tight budget.

More Team Members: The HR Agent will suggest when to hire. New team members just get a Developer account in the dashboard and can start working immediately.

More Agents: As you identify new needs, you can create custom agents through the dashboard. Just define a system prompt, select a model, and set approval rules.

Performance: If the app gets slow with many users, upgrade Hostinger to a higher Cloud plan with more RAM and CPU. The application architecture supports horizontal scaling.

Client Self-Service: As you grow, enhance the client portal to allow clients to request changes, view analytics, and manage their subscriptions without needing human intervention.

10.4 What If Something Goes Wrong?

API key runs out: The auto-failover system switches to the next key. You get an email alert. Just add a new key in the API Key Management page.

Agent gives bad output: Reject it in the approval queue. Add feedback explaining what was wrong. The agent will learn from the feedback in future interactions.

Dashboard goes down: Check Hostinger hPanel for error logs. Usually a restart fixes it. Your data is safe in the MySQL database.

Code generation is poor: Try being more specific in your instructions to the Dev Agent. Include examples, reference existing websites, or upload wireframes/screenshots.

Cline generates broken code: Tell Cline the specific error message. It will fix it. If it keeps failing, Kiran can step in and manually fix the issue.